THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Full Syllabus (Chapters 1–11) | Class 7 Science | Maximum Marks: 50

MULTIPLE CHOICE QUESTIONS — Answers — 50 × 1 = 50

Q.	Correct Option	Brief Reasoning
1	**(d)** an acid	Acid.
2	**(b)** glow dimly	4.5 V is less than the rated 6 V.
3	**(c)** magnesium	Basic oxide, so metal.
4	**(d)** a reversible chemical change	Hydrated and anhydrous forms.
5	**(c)** growth rates vary and the cousin may have started his spurt	Individuals grow differently.
6	**(a)** A is a better conductor	Faster fall means faster conduction.
7	**(d)** 20 m/s	(150+250)/20.
8	**(c)** fat-rich food	Bile aids fat digestion.
9	**(b)** the covered half remains brown with iodine	Light needed.
10	**(d)** returns along the same path	For 90°, reversed.
11	**(c)** more extreme	Larger variation.
12	**(a)** 40 mL	Proportional.
13	**(d)** test with a tester; add salt to water to make it conduct	Dissolved salts supply ions.
14	**(d)** magnesium	Magnesium is highly reactive.
15	**(c)** formation of a new substance	New substance.
16	**(d)** larynx is growing and adjusting	Settles in time.
17	**(a)** reversed	Land breeze.
18	**(b)** 15 km	60 - 45.
19	**(c)** 1,15,200 times	80 x 60 x 24.
20	**(a)** light energy to chemical energy	Stored in glucose.
21	**(b)** umbra	Complete shadow.
22	**(d)** more frequent leap years	Larger quarter.
23	**(a)** acidic	Acid rain.
24	**(b)** a conductor	Current passes through it.
25	**(b)** zinc	Acid reacts.
26	**(b)** store in dry place or oil them	Keeps moisture away.
27	**(c)** a rapid rise during the growth spurt	Sudden increase.
28	**(a)** sea breeze	Fluid motion.
29	**(b)** 1.82 s	Mean 18.2 / 10.
30	**(d)** to provide a large surface area for absorption	Large area.
31	**(c)** nitrogen-fixing bacteria	Rhizobium.
32	**(a)** virtual image position follows the law of reflection	Geometric consequence.
33	**(b)** the Earth always has a shadow but the Moon enters it only during eclipses	Shadow is always there.
34	**(a)** an acid	An acid neutralises the base.
35	**(c)** glow very bright and may fuse	Exceeds rating.
36	**(a)** sulphur	Non-metal.
37	**(a)** copper sulphate	Copper deposits.
38	**(b)** it is within the normal range	Normal variation.

39	**(b)** it absorbs more radiation	Dark absorbs.
40	**(a)** 300 m	300/30.
41	**(a)** Both A and R are true and R is the correct explanation of A.	Turmeric turns red (reddish brown) with bases, and soap solution is a base, so R explains A.
42	**(a)** Both A and R are true and R is the correct explanation of A.	Current flows only in a complete (closed) circuit, so R explains why the bulb glows.
43	**(a)** Both A and R are true and R is the correct explanation of A.	Malleability is the property of being beaten into thin sheets, so R explains A.
44	**(a)** Both A and R are true and R is the correct explanation of A.	Only the state changes from solid to liquid, so R explains why it is physical.
45	**(a)** Both A and R are true and R is the correct explanation of A.	Growth spurts and other changes of the teenage years are brought about by hormones.
46	**(a)** Both A and R are true and R is the correct explanation of A.	R describes the result of the flow from hot to cold, supporting A.
47	**(b)** Both A and R are true but R is not the correct explanation of A.	Both are true, but the relation between minute and second does not explain why the second is the SI unit.
48	**(a)** Both A and R are true and R is the correct explanation of A.	The enzyme in saliva starts acting on starch in the mouth, so R explains A.
49	**(a)** Both A and R are true and R is the correct explanation of A.	The green pigment traps light energy, which explains why chlorophyll is needed.
50	**(a)** Both A and R are true and R is the correct explanation of A.	Because light cannot bend around the opaque object, a dark region (shadow) forms behind it.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Full Syllabus (Chapters 1–11) | Class 7 Science | Maximum Marks: 50

MULTIPLE CHOICE QUESTIONS — Answers — 50 × 1 = 50

Q.	Correct Option	Brief Reasoning
1	**(d)** 360 km	Gap 120 km, closing 30 km/h in 4 h.
2	**(b)** 8000 mL	16 x 500.
3	**(d)** chemical energy to usable energy	Released.
4	**(a)** shadow	Dark.
5	**(b)** the Earth moves around the Sun meanwhile	Extra two days to realign.
6	**(d)** it is a base used to relieve acidity	Carbonates react with acids.
7	**(b)** an insulator	Current does not pass.
8	**(a)** copper	Good conductor.
9	**(a)** apply oil and keep dry	Oiling.
10	**(b)** around 11 to 13 years	Growth spurt.
11	**(d)** heating iron rod	Solid transfer.
12	**(b)** less than true by about 0.2 s	Late start.
13	**(d)** plant food takes longer to digest	Cellulose.
14	**(d)** nitrogen left by pulses	Rhizobium.
15	**(a)** red	Same colour.
16	**(c)** compare dates of nearest distance with seasons	January is nearest.
17	**(c)** turns almost neutral	They neutralise.
18	**(c)** series	Single path.
19	**(b)** zinc	Zinc.
20	**(b)** carbon dioxide	Lime water test.
21	**(a)** anaemia	Iron lack.
22	**(c)** hot air rising pushes it	Convection currents.
23	**(b)** 4 m/s	400/100.
24	**(b)** poor protein digestion and more germs reaching the intestine	HCl needed.
25	**(b)** NaOH absorbs carbon dioxide	No CO2.
26	**(d)** 1.6 m tall	Half.
27	**(d)** 182 days	Half.
28	**(d)** 500 kg	200 x 2.5.
29	**(b)** copper conducts better	Brightness suggests current.
30	**(a)** blue to green	$CuSO_4$ is blue.
31	**(d)** unburnt carbon, a chemical change	New substance.
32	**(d)** be affected over time	Hormone loss.
33	**(b)** not form	No temperature difference.
34	**(a)** 12 km	x/4 + x/6 = 5 gives x = 12.
35	**(b)** 480 litres	8 L x 60.
36	**(c)** more glucose than used in respiration	Growth.
37	**(b)** double	Distance ratio 2:1.
38	**(d)** much nearer to the Earth	It appears equally large in the sky.
39	**(b)** a carbonate and an acid	CO2 is released.
40	**(a)** both conduct, copper better	Brightness shows current.

41	**(a)** Both A and R are true and R is the correct explanation of A.	Rotation turns each part of the Earth to and away from the Sun every 24 hours.
42	**(d)** A is false but R is true.	Acids turn blue litmus red and do not change red litmus; so A is false, but R is true.
43	**(c)** A is true but R is false.	A fuse is in series, so A is true; but fuse wire has a LOW melting point and melts when current is too large, so R is false.
44	**(b)** Both A and R are true but R is not the correct explanation of A.	Both statements are true, but being a form of carbon does not explain why graphite conducts.
45	**(b)** Both A and R are true but R is not the correct explanation of A.	Both are true, but being a metal does not explain the need for oxygen and water.
46	**(c)** A is true but R is false.	The pituitary controls other glands, so A is true; insulin is made by the pancreas, so R is false.
47	**(a)** Both A and R are true and R is the correct explanation of A.	Only radiation can travel through empty space, so R explains A.
48	**(a)** Both A and R are true and R is the correct explanation of A.	R restates the effect of length on the time period and explains A.
49	**(c)** A is true but R is false.	Bile helps in fat digestion, so A is true; but bile contains no enzymes (it only breaks fat into tiny droplets), so R is false.
50	**(b)** Both A and R are true but R is not the correct explanation of A.	Both are true, but guard cells do not explain why gases exchange occurs through stomata.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Full Syllabus (Chapters 1–11) | Class 7 Science | Maximum Marks: 50

MULTIPLE CHOICE QUESTIONS — Answers — 50 × 1 = 50

Q.	Correct Option	Brief Reasoning
1	**(c)** a single path for the current	The loop is broken.
2	**(c)** less reactive than copper	Copper displaces it.
3	**(a)** the wax changes into new substances such as carbon dioxide and water vapour	Mass is conserved.
4	**(a)** swelling of the neck	Goitre.
5	**(a)** radiation	Direct radiation.
6	**(c)** 200 m	Back at the start.
7	**(b)** poor absorption of digested food	Less surface area.
8	**(a)** increases with light intensity	Closer lamp.
9	**(b)** 12 m	2 x 6.
10	**(c)** overhead all year	No tilt.
11	**(b)** the acid was stronger or more concentrated	Excess acid.
12	**(d)** the nail is coated with paint or rust	Surface layer blocks contact.
13	**(b)** increases	Copper heavier.
14	**(c)** a chemical change	Energy released.
15	**(a)** rise	No control.
16	**(c)** heats and cools quickly	Low heat capacity.
17	**(d)** 40 km	8 km/h x 5.
18	**(d)** 10 million	5 x 2.
19	**(c)** it gains mass and grows	Balanced gas exchange.
20	**(a)** 50 cm	20 x 100/40.
21	**(a)** about 2.5 times	The Earth's shadow is large.
22	**(c)** acid-carbonate reaction	CO2.
23	**(d)** goes off	The loop is broken.
24	**(d)** less reactive	They do not displace hydrogen.
25	**(a)** silver reacts with sulphur in air	Alloy resists.
26	**(a)** common as cycles settle	Cycles take time.
27	**(d)** hot gases rise and draw fresh air	Draught by convection.
28	**(a)** 0.25 per cent	0.1/40.
29	**(b)** the iron-containing haemoglobin	Iron.
30	**(a)** saprotroph	Feeds on bread.
31	**(a)** light from a lamp reflecting from the book	Absorbs red.
32	**(d)** incorrect	Far side gets sunlight.
33	**(b)** $\mathrm{HCl} + \mathrm{NaOH} \longrightarrow \mathrm{NaCl} + \mathrm{H_2O}$	Neutralisation.
34	**(c)** its melting point is lower than that of the connecting wires	Low melting.
35	**(d)** carbon	CO2.
36	**(a)** a new black solid not attracted by a magnet	New compound.
37	**(d)** insulin – pancreas	Insulin is from the pancreas.
38	**(d)** white reflects most radiation	Less absorption.
39	**(a)** 30 km/h	2x/(x/20 + x/60) = 30.
40	**(b)** inhalation is affected	Breathing needs it.

41	**(b)** Both A and R are true but R is not the correct explanation of A.	Both are true; the Moon only reflects sunlight, but the luminous nature of the Sun does not explain it.
42	**(d)** A is false but R is true.	Seasons are due to the tilt of the axis and revolution, not distance, so A is false; R is true.
43	**(b)** Both A and R are true but R is not the correct explanation of A.	Both statements are true, but the release of heat does not explain why salt and water form.
44	**(b)** Both A and R are true but R is not the correct explanation of A.	Both are true, but the plastic covering is used because it is an insulator, which R does not explain.
45	**(a)** Both A and R are true and R is the correct explanation of A.	Iron takes the place of copper in the solution and copper is deposited on the nail.
46	**(a)** Both A and R are true and R is the correct explanation of A.	Melting is physical and burning is chemical, which R states and so explains A.
47	**(b)** Both A and R are true but R is not the correct explanation of A.	Both are true, but one fact does not explain the other.
48	**(b)** Both A and R are true but R is not the correct explanation of A.	Both are true, but one does not explain the other.
49	**(c)** A is true but R is false.	$90/2 = 45$ km/h so A is true; speed = distance / time, so R is false.
50	**(d)** A is false but R is true.	Arteries carry blood away from the heart, so A is false; R is true.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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